🔑 Grok API

Grok API Key (x.ai)

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✅ API key saved

⚙️ Settings

Signal Horizon

1mo

Grok Model

grok-4.20-reasoning

🔄 Refresh All Data & Signals

Not financial advice.

🌍 GeoSupply Short-Term Profit Predictor v4.0

Self-Improving • Persistent Memory • Sector Heatmap | Geo-Supply Chain Alpha

📈 Leaderboard 📊 Charts 🔥 Sector Heatmap 🤖 Grok Thesis 📜 Grok

🤖 Grok 2-5 Day Thesis

Generate Thesis + Self-Improvement Suggestions

Thesis: Critical Minerals + Energy Transition Momentum

The leaderboard reveals a concentrated **geo-supply chain thesis** centered on **copper/base metals** and **energy security**. Mining names (VALE, FCX, BHP.AX, FMG.AX, S32.AX) dominate alongside Energy (WDS.AX, STO.AX, COP, OXY), reflecting sustained tightness in commodities critical for electrification, AI infrastructure, and defense. Copper stands out as the clearest nexus — AI power demand + grid buildout + persistent supply risks (Chile/Peru labor, China stockpiles, Western reshoring). Iron ore names are riding secondary China stimulus hopes.

AMD is the lone Tech outlier but fits the broader theme: semiconductor supply chain security (CHIPS Act, AI capex). Volume spikes across the list (0.73–0.84) confirm institutional rotation into real assets and enabling technologies. RSI dispersion is useful — overbought miners (VALE 76, FCX 78) suggest momentum but caution, while COP (45) offers cleaner risk/reward.

2-5 Day Trade Ideas (short-term momentum/mean reversion)

1. AMD Long (Highest Conviction)

- **Rationale**: Clear #1 signal (12.97 score), 15.5% 1mo move, strong vol spike (0.81), RSI 67.9 still has room. AI capex narrative remains unstoppable and maps directly to geo-supply (Taiwan risk, US onshoring).

- **Setup**: Long stock or Aug 15/20 calls. Add on dips toward $228–232.
- **Target**: $248–252 (5–6% upside).
- **Stop**: $225. Expected hold 2–4 days.

2. FCX Long (Copper Supply Chain Premium)

- **Rationale**: Pure-play copper with strong signal (5.64), elevated vol, and sector tailwinds. Copper's dual use in data centers + renewables creates persistent bid against fragmented supply.
- **Setup**: Long above `66.50 or on pullback to` 64.80.
- **Target**: $71–72 (quick 7–8% move).
- **Stop**: $63.80. Watch for continuation if it takes out recent highs.

3. COP Long (Best Risk/Reward Setup)

- **Rationale**: Lowest RSI (45) among leaders yet strong vol spike (0.84) and clean energy security signal. Less extended than the miners, better short-term entry as oil finds support amid geopolitical noise.
- **Setup**: Long $122–123 zone.
- **Target**: $129–130.
- **Stop**: $119.50. 3–5 day swing.

Risk Note: Miners with RSI >75 (VALE, FCX) are prone to sharp reversals — size accordingly. Focus flow on AMD and COP for the cleanest 2–5 day setups. Monitor copper futures and China stimulus headlines as catalysts.

💡 Self-Improvement Suggestions (for next version)

3 Concrete Code/Architecture Improvements for GeoSupply v4.1

1. Introduce a Thematic Analysis Pipeline (Strategy + Composite Pattern)

Current Problem: The thesis generation logic appears to be a single large prompt/model call that hard-codes the "Critical Minerals + Energy Transition" narrative. This will become unmaintainable as you add Lithium, Rare Earths, Uranium, and Defense-critical materials.

Proposed Architecture:

```
# v4.1 structure
thematic_pipeline = ThesisPipeline([
    CopperNexusAnalyzer(weight=0.35),
    EnergyTransitionAnalyzer(weight=0.25),
    SupplyRiskAnalyzer(weight=0.20),
    VolumeMomentumAnalyzer(weight=0.12),
    CrossSectorContagionAnalyzer(weight=0.08),  #
])

thesis = thematic_pipeline.analyze(leaderboard_sn
```

Benefits:

- Each analyzer returns structured signals (`Signal` objects with confidence, evidence, entities) instead of free text.
- Easy to A/B test or swap analyzers.
- The final `ThesisComposer` only does narrative synthesis from structured signals (much more controllable).

Implementation effort: 2–3 days. High ROI.

2. Replace Flat Leaderboard with a Supply Chain Graph Layer (Knowledge Graph + Vector Hybrid)

Current Problem: You're losing enormous value by treating the leaderboard as a flat list. The real alpha is in the *relationships* (FCX → Chile labor → AI power demand → AMD).

Recommended Stack for v4.1:

- **Neo4j** (or Memgraph) as source of truth for the supply chain ontology:
  - Nodes: `Company` , `Commodity` , `Country` , `RiskEvent` , `EndUse` (AI DataCenter, Grid, EV, Defense)
  - Relationships: `SUPPLIES` , `EXPOSED_TO` , `ENABLES` , `RISK_IMPACTS`
- **Qdrant/Pinecone** vector store containing embeddings of every node + previous high-quality theses.

- Materialized leaderboard becomes a *graph query* + vector similarity pass.

Example query:

```
MATCH (c:Commodity {name: "Copper"})-[:ENABLES]->
MATCH (c)<-[:SUPPLIES]-(company:Company)
RETURN company.ticker,
       calculate_geopolitical_risk(company, $risk
       use.demand_growth as thesis_momentum
```

This turns your app from "another stock screener" into the definitive geo-supply intelligence platform.

3. Implement Structured Thesis Output + Versioned Thesis Store (Schema-First Generation)

Current Problem: The thesis is free-form text. You cannot reliably track which theses performed well, which signals were predictive, or build a feedback loop.

Solution:

```
class GeoSupplyThesis(BaseModel):
    core_narrative: str
    primary_commodity: str
    secondary_commodity: Optional[str]
    key_drivers: List[Driver]           # Driver
    implicated_tickers: List[TickerSignal]
    conviction_score: float             # 0.0-1.0
    time_horizon: Literal["1Q", "2-3Q", "2025-202
    version: str = "4.1"
    generated_at: datetime
    evidence_nodes: List[str]           # referen
```

Architecture:

- All model calls must return Pydantic-validated structured output (use Outlines, Guidance, or Instructor).
- Every generated thesis is stored in a `thesis_history` table/graph with its performance metadata (subsequent 1W/4W/12W returns of implicated names).
- This creates a self-improving flywheel for v5.

Priority Recommendation: Implement **#1 (Thematic Pipeline)** immediately — it unblocks the other two and gives you the cleanest win for v4.1. Then tackle the Knowledge Graph layer (#2), which is your real defensibility moat.

Would you like me to provide the actual folder structure + key class skeletons for this new architecture?

Repo: https://github.com/JeffStone69/XAi • v4.0 Merged